CESI
ÉCOLE D'INGÉNIEURS

INTERNSHIP PROGRAM ABROAD – SUBJECT VALIDATION FORM
CESI SCHOOL OF ENGINEERING

Specify if this internship is part of the end-of-studies project: no

General information	
Student	Family name: Silici—Sieng Kenzo Status: Student Major : - Computer science
Hosting organization	Name of the hosting organization: Innovaphone AG Address: Umberto-Nobile-Straße 15 Postal code: 71063 City: Sindelfingen Country: Germany Telephone: +49(7031)73009-0 Number of employees: 150 Main activity: VoIP, UCC
Internship supervisor	Full name and position: Norbert Diehl Direct line and/or mobile: Email: ndiehl@innovaphone.com
Contract information	Legal signatory of the internship contract (name and position): Legal representative of the organization (name and position): Administrative contact in the organization (name, telephone, email): Kim-Moana Hofmeister Address where the contract is to be sent: Umberto-Nobile-Straße 15 | 71063 Sindelfingen | Germany Location of the internship: Umberto-Nobile-Straße 15 | 71063 Sindelfingen | Germany

	Start date: 13/04/2026 End date: 3/07/2026 Proposed payment: 14€ per hour
Safety of the destination	On the website of the French Ministry of Foreign Affairs (Travel Advice section), how is the destination of your international internship classified: - Level red: prohibited by the school - Level orange: prohibited by the school - Level yellow: reinforced vigilance - Green level: normal vigilance Please attach a copy (screenshot) of the map:

CESI
ÉCOLE D'INGÉNIEURS

Internship details

Please provide as much information as possible, especially if the internship is part of the End-of-studies project, as this will provide the basis for the student's project management plan (see at the top of the document).

Initial situation/context

The organization's current software portfolio is built around a high-performance SDK with an extensive codebase written in C++. At present, app services utilizing this SDK are developed using either C++ or JavaScript. To modernize the development environment, the company intends to introduce Rust as an additional supported language for both new SDK components and app services.

Stakes

To ensure a stable transition without disrupting existing operations, the following requirements apply.

- **Dual-language coexistence: Rust is intended as a supplemental option for new development. There is no immediate requirement to port or rewrite the existing C++ codebase.**
- **Maintenance: Existing C++ products will remain in active maintenance, requiring the SDK to support heterogeneous environments.**
- **Interoperability: A "smooth transition" is defined by the ability of new Rust code to interface directly with the existing C++ SDK and vice versa.**

Objective

The primary objective of this project is to establish a technical foundation for integrating Rust into the company's existing software ecosystem. This involves transitioning from an analytical understanding of the current architecture to the practical creation of standardized development templates.

Tasks or missions entrusted to the intern

- **Analysis of the current C++ codebase and identifying the typical patterns used in app services.**
- **Research of technical solutions for interfacing between Rust and C++.**
- **Development of an app service template written in Rust, that uses functionality of the existing C++ SDK.**
- **Development of an SDK library stub written in Rust that can be used from both Rust and C++ code.**

Persons or departments involved in the projects

Developing team

Norbert DIEHL (Developer and internship supervisor)

Matthias SCHERTLER (Head of Development team)

Available resources

Actual codebase

Additional resources to be provided

CESI
ÉCOLE D'INGÉNIEURS

Developing team feedback and experiences

Main tasks	Provisional dates	Duration
Learning about Rust, analyzing existing C++ codebase, tooling and FFI research, <Hello World> PoC project	**13/04 – 01/05**	**3 weeks**
Rust template for app service with bindings for core C++ functions, documentation and abstraction of C++ functions into Rust equivalents	**04/05 – 22/05**	**3 weeks**
SDK library stub: Rust module as shared library, C-exports and header files usable by C++ SDK, testing the integration and validation of memory usage (ownership concept vs. raw pointers)	**25/05 – 19/06**	**4 weeks**
Quality assurance and final report/presentation: testing and benchmarking (original C++ functions vs. Rust integrations), summary and presentation of research, choices, testing and a working demo project	**22/06 – 3/07**	**2 weeks**

Deliverables and due dates

Documentation of research and choices regarding the tooling used for code integration, C++ codebase (common patterns) and their Rust integrations

App service template in Rust & SDK library stub

Benchmarking results (original C++ vs. Rust integration) & Demo project utilizing all developed components

Signatures

After completion and validation of this document, the internship contract will be drafted and sent to all concerned parties.	Student : Kenzo SILICI--SIENG (Name and position)	Date and stamp :16/03/2026
	~~In case of apprenticeship, French employer:~~ ~~(Name and position)~~	~~Date and stamp:~~
	Hosting organization: Dirk Meyer (Name and position) Head of Operations	Date and stamp: 16/03/2026
	CESI School of Engineering: (Name and position)	Date and stamp: